

Brain-Computer Interaction using OpenViBE

JAI SACHITH PAUL

Brain signals communicating directly to a computer is never an easy task, yet it is interesting that it could help humanity in a multitude of ways. In this article, we will discuss OpenViBE software platform for the design, test and use of brain computer interfaces (BCI).

A set of software modules in OpenViBE takes input from the acquisition machine [generally, electro-encephalography (EEG) signals] connected to the head, measuring electrical activity on the scalp that roughly reflects the activity of cortical neuron population.

Communication engineers around the world make use of OpenViBE for real-time processing of brain signals. The software is used to acquire, filter, process, classify and visualise brain signals in real-time. This tool finds applications in various fields including medical (assistance to disabled people, real-time biofeedback, neuro feedback and real-time diagnosis), multimedia (virtual reality, or VR, and video games), robotics and all other application fields related to brain-computer interfaces and real-time neurosciences.

The software works well on Windows and Linux operating systems and is released under GNU Affero general-public licence v3.0 (AGPL-3). You can find the latest version of this environment, OpenViBE 1.0.0, in this month's DVD for EFY Plus.

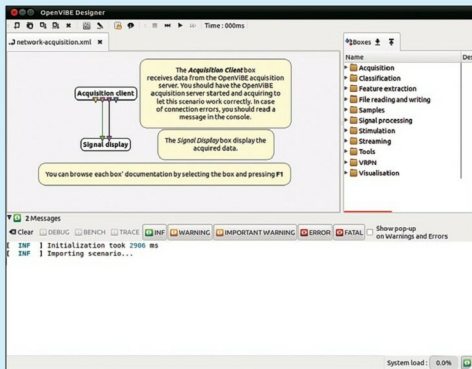


Fig. 1: OpenViBE Designer with tabbed interface and graphical language

Real-time 3D virtual brain visualisation

In neuro feedback and BCI systems, the target region in the brain is continuously monitored to extract features from it. The participant gets back this information in the form of visual and auditory objects, where one or more characteristics co-vary in real-time with the extracted feature. With 2D modelling techniques it is possible to extract a maximum of only three features in real-time. Neither can we extract features from different areas of the brain nor can we have a global view of the same.

OpenViBE software makes use of real-time 3D virtual brain visualisation to make up for this limitation. It makes use of 3D-functional electromagnetic data to represent brain activity in a realistic 3D brain model. EEG of the participant is converted into intracranial current density, which conserves real spatial and temporal relations of the signal as much as possible. The participant can virtually navigate into his or her brain and watch its electromagnetic dynamics. Since this technique enhances a holistic form of data representation, it is very useful for EEG data analysis too.

Who can make use of OpenViBE

The software is designed to cater to the needs of mainly four types

of users—developers and application developers, who are basically programmers, can make use of the scripting language, while authors